

HADAMARD DOMAIN TRAINING WITH INTEGERS FOR CLASS INCREMENTAL QUANTIZED LEARNING

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ABSTRACT

Continual learning is a desirable feature in many modern machine learning applications, which allows in-field adaptation and updating, ranging from accommodating distribution shift, to fine-tuning, and to learning new tasks. For applications with privacy and low latency requirements, the compute and memory demands imposed by continual learning can be cost-prohibitive for resource-constrained edge platforms. Reducing computational precision through fully quantized training (FQT) simultaneously reduces memory footprint and increases compute efficiency for both training and inference. However, aggressive quantization especially integer FQT typically degrades model accuracy to unacceptable levels. In this paper, we propose a technique that leverages inexpensive Hadamard transforms to enable low-precision training with only integer matrix multiplications. We further determine which tensors need stochastic rounding and propose tiled matrix multiplication to enable low-bit width accumulators. We demonstrate the effectiveness of our technique on several human activity recognition datasets and CIFAR100 in a class incremental learning setting. We achieve less than 0.5% and 3% accuracy degradation while we quantize all matrix multiplications inputs down to 4-bits with 8-bit accumulators.

1 INTRODUCTION

Humans have a remarkable ability to continuously acquire new information throughout their lives. Given neural networks (NN) this capability can make them more versatile and useful, as tasks can change over time, and data distributions can shift. Even when a task remains constant, the integration of new data is crucial for improving predictive outcomes. To accomplish this, NNs must strike a delicate balance, being adaptable enough to accommodate new tasks and data, while also remaining stable enough to preserve previously acquired knowledge. This inherent tension between adaptability and stability is referred to as the "Stability-Plasticity Dilemma" (Mermillod et al., 2013), and is the subject of Continual Learning (CL), Lifelong Learning, or Incremental Learning. A neural networks' inability to acquire new tasks without compromising their performance on previously learned tasks is commonly referred to as "Catastrophic Forgetting" (CF). Addressing CF stands as a significant obstacle in the pursuit of more capable autonomous agents, as highlighted by McCloskey & Cohen (1989); Zhou et al. (2023). Techniques to mitigate the impact of CF include preventing information loss through regularized parameter updates, memory replay, or dynamic architectures (Parisi et al., 2018). Harun et al. (2023a) found that many popular continual learning algorithms are resource-expensive, with some CL algorithms being more costly than training the same model from scratch. These costs are further exacerbated when additional constraints such as privacy, low-latency, or low-energy are imposed.

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